

# AJAY VENKATESH

Brooklyn, NY

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## Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

## Experience

### • Campus Mesh

Jan 2026 – Present

AI Software Engineer

New York, US

- Built **CampusIQ**, an LLM-powered AI assistant, with iterative **prompt engineering** and **RAG retrieval** over structured data and **embedding-based semantic search**, improving contextual accuracy by **30%**.
- Designed a personalized **recommendation engine** using collaborative filtering on user interaction signals, increasing engagement by **28%** and retention by **50%**.
- Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management.

### • Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
- Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
- Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.

### • New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

AI Software Engineer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
- Built end-to-end **computer vision pipelines (YOLOv8, EfficientNet)** for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
- Developed a full-stack insurance analytics product (**React, Node.js, REST APIs**) integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with control; processed **250K+** records and powered **AWS QuickSight** dashboards.

### • PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Designed a distributed backend on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting **ML inference pipelines** for healthcare analytics.
- Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate document data extraction, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** on **100K+** records, reducing execution from **5 hours to 1 hour**.
- Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

## Projects

### • InviLite (AI-Driven Data Engineering & Analytics Platform) | AWS, SageMaker, Bedrock, FAISS, Python

- Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.

### • Hospital Management System | React, Node.js, Express, MongoDB, SageMaker, Hugging Face

- Built a full-stack healthcare platform (React, Node.js, Express, MongoDB) managing appointments, patient records, and lab reports with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker** for real-time diagnostic support.

## Skills

- **Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Computer Vision (YOLOv8, EfficientNet), Time-Series Modeling, MLflow, SVM, Random Forest, Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Cursor, GitHub Copilot, Model Context Protocol (MCP), 3D point cloud processing, Open3D, PointNet++, Transformer-based architectures, Graph-based architectures, Distributed training, Hyperparameter optimization, Sensor fusion, Kalman filtering

## Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

## Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals